

# Team Contributions

The Multifunctional Fireguard Bot project was successfully completed through the combined efforts of all team members. Each member contributed significantly to different aspects of the project, from hardware implementation to software development and documentation.

| Name               | Student ID | Detailed Contribution                                                                                                                                                                                                                                                                                                                                                                                      |
|--------------------|------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Osman Goni         | 23101012   | Developed the core Arduino Uno firmware including sensor reading, line following logic, and fire detection system. Implemented communication protocol between ESP32-CAM and Arduino Uno. Actively participated in hardware assembly, wiring, sensor calibration, and mechanical integration. Contributed to system testing and troubleshooting.                                                            |
| Aeysha Tabassum    | 23101014   | Configured ESP32-CAM pins, camera settings, and motor driver connections. Designed and developed the complete responsive Web User Interface (HTML, CSS & JavaScript). Implemented live camera streaming and tab-based mode control system. Contributed to hardware component selection, circuit design, and physical assembly of the robot. Assisted in documentation and presentation slides preparation. |
| Arannamoy Mondal   | 23101024   | Developed the complete YOLOv11-based AI navigation system in Python with real-time obstacle detection and decision-making logic. Implemented advanced autonomous control features. Contributed to Arduino firmware development for Firefighter mode. Participated in hardware assembly and performed extensive testing of autonomous and AI modes.                                                         |
| Fabia Akter Borsha | 23101025   | Developed motor control functions and PWM setup. Implemented all web server endpoints (handleCommand(), handleMode(), handleSpeed(), handleEmergency()). Designed fail-safe logic and Arduino serial communication handling. Contributed to hardware assembly, wiring, and overall system integration. Assisted in system testing and debugging.                                                           |

## Collective Contribution

All team members collectively participated in hardware component procurement, robot chassis assembly, wiring, sensor calibration, system integration, testing, and evaluation. The team also jointly prepared the project report, presentation slides, and final documentation including system testing, evaluation, and result analysis.